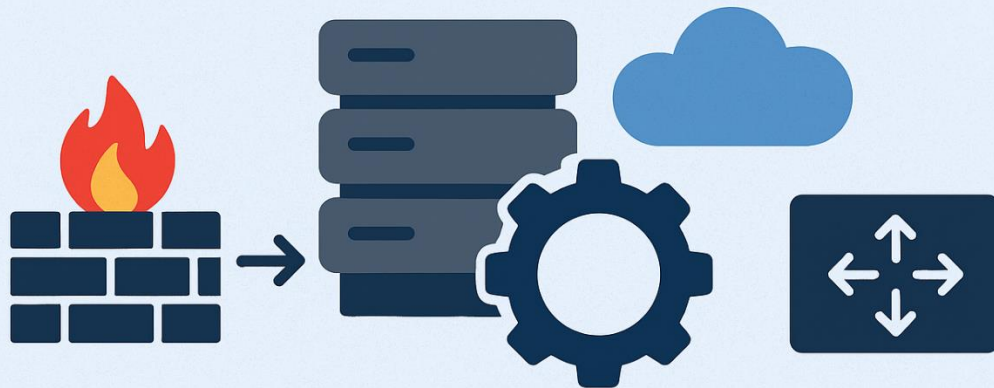


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OpenShift UPI Installation – Prerequisites for On-Prem Environment

OpenShift UPI Installation Prerequisites for On-Prem Environment



Introduction

Deploying Red Hat OpenShift using the **User-Provisioned Infrastructure (UPI)** model gives full control over the infrastructure setup.

Before installation, the environment must be properly prepared with the right **hardware, networking, storage, DNS, and security configurations**.

This document lists the **key prerequisites** required to host OpenShift on an **on-premises data center**. It is **not an installation guide**, but a readiness checklist for infrastructure and system administrators.

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Infrastructure and Hardware Requirements

Node Requirements

Each node type requires specific resources for stable operation.

Node Type	vCPU	RAM	Storage	Description
Bootstrap	4	16 GB	120 GB	Temporary node for cluster bring-up
Master	4	16 GB	120 GB	Runs API server, scheduler, etcd
Worker	2	8 GB	120 GB	Runs workloads and applications
Infra Node (optional)	4	16 GB	120 GB	Handles routing, registry, monitoring

- All nodes must support **64-bit architecture** and **virtualization extensions** (VT-x/AMD-V).
 - **Time synchronization** via NTP or Chrony is mandatory across all nodes.
 - Ensure **redundant power supply and cooling** in the data center for continuous uptime.
-

Networking and DNS Requirements

IP Addressing

- Assign **static IPs** to all nodes.
- Reserve additional IPs for:
 - API endpoint
 - API internal endpoint
 - Ingress endpoint

Example:

api.ocp.example.com → 10.10.0.5
api-int.ocp.example.com → 10.10.0.6
*.apps.ocp.example.com → 10.10.0.7

DNS Configuration

DNS resolution must be in place before installation.

Required records:

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- `api.<cluster>.<domain>` → Public API endpoint
- `api-int.<cluster>.<domain>` → Internal API endpoint
- `*.apps.<cluster>.<domain>` → Routes via Ingress
- Node hostnames and reverse DNS records

Load Balancer Setup

Two load balancers are required:

1. **API Load Balancer** – For control plane traffic (ports 6443, 22623).
2. **Ingress Load Balancer** – For applications (ports 80, 443).

Ensure the load balancer supports **TCP passthrough** and **health checks**.

Firewall and Port Requirements

Allow the following:

- TCP 6443 – API server
- TCP 22623 – Machine config
- TCP 2379–2380 – etcd
- TCP/UDP 4789 – VXLAN (SDN)
- TCP 80, 443 – Web traffic
- TCP 10250 – Kubelet communication

Storage and Registry Prerequisites

Storage Setup

Persistent storage is critical for OpenShift components such as registry, logging, and monitoring.

Supported on-prem storage options:

- NFS
- iSCSI
- Fibre Channel
- Ceph / OpenShift Data Foundation
- Local PV (for test)

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Provision high-performance block storage for workloads requiring low latency (e.g., databases).

Registry Storage

The internal image registry requires **at least 100 GB** of persistent storage. Use shared and fault-tolerant storage like NFS or ODF for production.

Default Storage Class

Before workload deployment, define a **default storage class** (e.g., provisioner for dynamic PVs).

Operating System and Access Requirements

Supported OS

- **RHCOS** is used for cluster nodes.
- **RHEL 8 or 9** can be used for bastion or utility servers (DNS, NTP, HTTPD). Ensure valid **Red Hat subscription** and access to necessary repositories.

User Access

- The provisioning user must have **sudo/root** privileges.
- Enable **passwordless SSH** between bastion and all nodes for automation.

Tools on Bastion Host

Install the following packages and utilities:

wget, curl, httpd, tar, unzip, jq, podman, nmcli

Download required OpenShift tools:

- openshift-install
 - oc (OpenShift CLI)
 - kubectl
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Cluster Configuration and Security Requirements

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Pull Secret

A valid **Red Hat pull secret** is needed to access image registries:

<https://cloud.redhat.com/openshift/install/pull-secret>

SSH Keys

Generate SSH keys and include the public key during cluster setup to enable secure access.

Certificates

- Ensure DNS names match certificates.
- Use **trusted CA-signed certificates** for production.

Security Controls

- Enable **SELinux in enforcing mode**.
- Restrict ports to only what's needed.
- Configure **RBAC roles** post-installation for controlled access.
- Maintain audit logs for node and API activities.

Validation and Readiness Checklist

Before proceeding with OpenShift installation, confirm the following:

Category	Check
Hardware	Sufficient CPU, RAM, storage for each node
Networking	Static IPs, DNS resolution verified
Load Balancer	Configured with correct ports and health checks
Storage	Persistent and default storage class ready
Access	SSH, NTP, and subscriptions active
Security	SELinux, firewall, certificates validated

Summary

Preparing an on-prem environment for OpenShift UPI deployment demands careful **planning, validation, and documentation**.

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Meeting these prerequisites ensures:

- Stable control plane and workload operation
- Seamless communication between components
- Reliable and scalable cluster performance

Thorough pre-installation checks help prevent issues during deployment and reduce maintenance overhead in production environments.

Key Points to Remember

- UPI offers flexibility but requires precise setup.
- Validate DNS, IPs, and certificates before running the installer.
- Keep infrastructure diagrams and IP tables for reference.
- Maintain consistent OS versions and synchronized time across nodes.

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